

Question 1

(a) Differentiate:

(i) e^{3x-1}

1

(ii) $\log_e (2x - 1)$

1

(b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

1

(c) Find the remainder when the polynomial $P(x) = x^3 - 2x$ is divided by $x + 1$

2

(d) Evaluate $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$

2

(e) Find the equation of the tangent to the curve $y = e^{2x} - 3x$ at the point $(0,1)$

2

(f) $A(-2, -5)$ and $B(1, 4)$ are two points. Find the acute angle θ between the line AB and the line $x + 2y + 1 = 0$, giving the answer correct to the nearest minute.

3

Question 2

Start a new page

(a) Given $\log_a x = 0.64$ and $\log_a y = 0.04$ find $\log_a \left(\frac{x}{y} \right)$ 1

(b) Q(-1, 4) and R(x, y) are two points. The point P(14, -6) divides the interval QR externally in the ratio 5:3. Find the coordinates of R. 2

(c) (i) Sketch $y = e^x - 1$ 2

(ii) Find the exact volume of the solid of revolution formed when the curve $y = e^x - 1$ is rotated about the x-axis from $x = 0$ to $x = 1$ 3

(d) (i) Show that the function $x^3 - 2x - 5 = 0$ has one root which lies between 2 and 2.2 1

(ii) Using $x = 2.1$ as an approximation to a root of $x^3 - 2x - 5 = 0$, find a better approximation, correct to 2 decimal places, using Newton's method once. 3

$\frac{1}{1} e^x$

Question 3**Start a new page**

(a) Find $\int \frac{x}{x^2 + 5} dx$ **2**

(b) Write the expansion of $\sin (A + B)$.
Hence, or otherwise, find the exact value of $\sin 75^\circ$ **3**

(c) Solve for x : $\frac{4}{x+1} \leq 3$ **3**

(d)

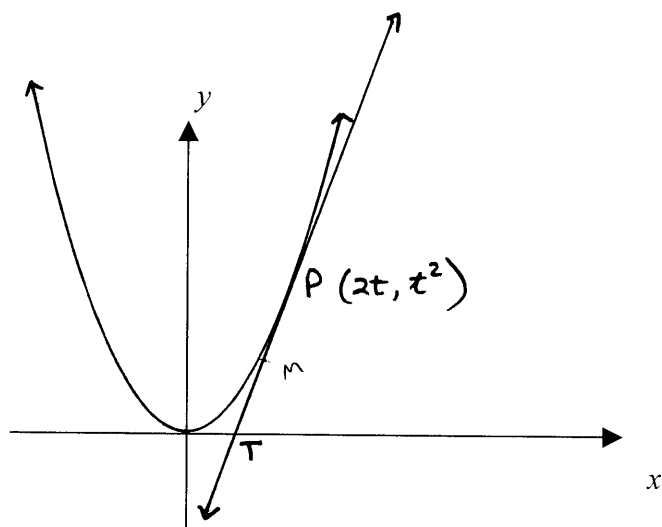
(i) Show that $(x + 1)$ is a factor of $f(x) = 2x^3 + 7x^2 - 7x - 12$ **1**

(ii) Find all roots of $f(x) = 2x^3 + 7x^2 - 7x - 12$ and hence sketch the curve. **3**

(a) If α, β, γ are the roots of the equation $3x^3 - 6x^2 + x + 2 = 0$ find the value of

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

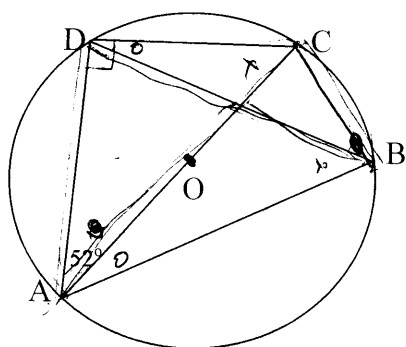
(b)



P ($2t, t^2$) is a variable point which moves on the parabola $x^2 = 4y$. The tangent to the parabola at P cuts the x-axis at T. M is the midpoint of PT.

- (i) Show that the tangent PT has equation $tx - y - t^2 = 0$ 2
- (ii) Show that M has coordinates $(\frac{3t}{2}, \frac{t^2}{2})$ 2
- (iii) Hence find the Cartesian equation of the locus of M as P moves on the parabola 2

(c)



not to scale

Circle, centre O, has diameter AC. $\angle DAC = 52^\circ$

- (i) Explain why $\angle ADC = 90^\circ$
- (ii) Find $\angle DBA$, giving reasons for your answer.

Question 5

Start a new page

(a) Stephanie invests \$150 at the start of each month into a superannuation fund. The interest is compounded monthly at a rate of 3% p.a. The first \$150 is invested at the beginning of January 2005 and the last is invested at the beginning of December 2010. Calculate, to the nearest dollar:

- | | | |
|------|---|---|
| (i) | The amount to which the January 2005 investment will have grown by the end of 2010. | 2 |
| (ii) | The amount to which the total will have grown by the end of 2010. | 3 |

(b) Use mathematical induction to prove that:

$$\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \dots + \frac{1}{(n+1)(n+2)} = \frac{n}{2(n+2)} \quad 3$$

(c) Consider the function $f(x) = 2 \sin^{-1}x$

- | | | |
|-------|---|---|
| (i) | Sketch the graph of $f(x) = 2 \sin^{-1}x$ | 1 |
| (ii) | Find the exact value of $f\left(\frac{1}{\sqrt{2}}\right)$ | 1 |
| (iii) | Find the equation of the tangent to the curve at the point where $x = \frac{1}{\sqrt{2}}$ | 2 |

Question 6

Start a new page

(a) Find $\frac{d}{dx} \cos^{-1} 2x^3$

2

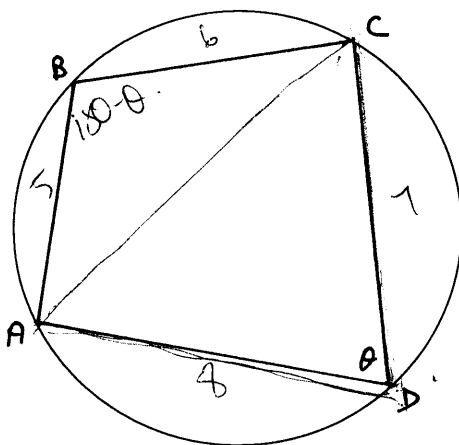
(b) If $t = \tan \frac{1}{2} \theta$, prove $\cos \theta (\tan \theta - \tan \frac{1}{2} \theta) = \tan \frac{1}{2} \theta$

3

(c) Show that $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x dx = \frac{\pi}{8} - \frac{1}{4}$

3

(d)



A, B, C, D are points on the circumference of a circle.

AB = 5cm, BC = 6cm, DC = 7cm, AD = 8cm

(i) If $\angle ADC = \theta$, explain why $\angle ABC = 180^\circ - \theta$

1

(ii) By drawing the diagonal AC, or otherwise
 $\cos \angle ADC = \frac{13}{43}$

3

Question 7
Start a new page

- (a) Using the substitution $u = e^x$, find

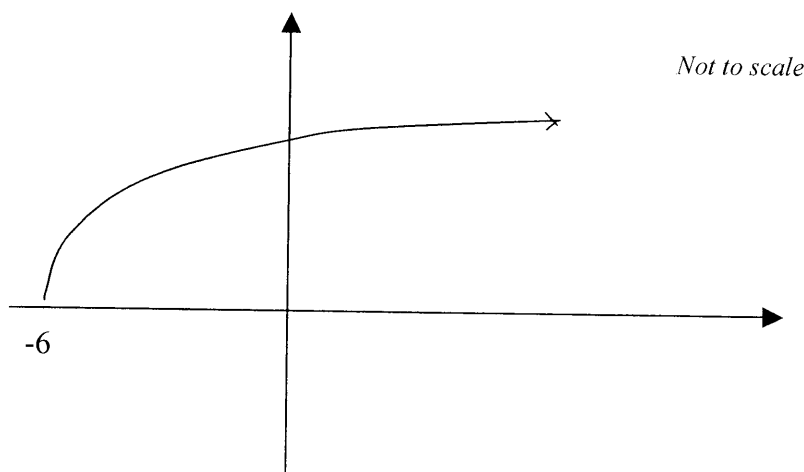
$$\int \frac{e^x}{1 + e^{2x}} dx \quad 2$$

(b)

- (i) Express $\sqrt{3} \sin \theta - \cos \theta$ in the form $R \sin (\theta - \alpha)$, where α is in radians. 2

- (ii) Hence, or otherwise, find all angles θ , where $0 \leq \theta \leq 2\pi$, for which $\sqrt{3} \sin \theta - \cos \theta = 1$ 2

- (c) The graph of $f(x) = \sqrt{x+6}$ for $x \geq -6$ is shown in the diagram.



- (i) Find the inverse function $f^{-1}(x)$ 1
- (ii) On the same diagram sketch the graphs of $y = f(x)$, the line $y = x$, and $y = f^{-1}(x)$. Show clearly the intercepts on the coordinate axes. 2
- (iii) What is the domain of $f^{-1}(x)$ 1
- (iv) Show that the x coordinate of any points of intersection of the graphs $y = f(x)$ and $y = f^{-1}(x)$ satisfy the equation $x^2 - x - 6 = 0$. Hence find any points of intersection of the two graphs. 2

END OF EXAMINATION

QUESTION 1

$$a) i) y = e^{3x-1}$$

$$y' = 3e^{3x-1}$$

$$ii) y = \log_e(2x-1)$$

$$y' = \frac{2}{2x-1}$$

$$b) \lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{3}{1}$$

$$= 3$$

$$c) P(x) = x^3 - 2x$$

$$P(-1) = (-1)^3 - 2(-1)$$

$$= -1 + 2$$

$$= 1$$

$$d) \int_0^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}}$$

$$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0$$

$$= \frac{\pi}{3} - 0$$

$$= \frac{\pi}{3}$$

$$e) y = e^{2x-3}$$

$$y' = 2e^{2x-3}$$

$$\text{at } x=0, y' = 2e^0 - 3$$

$$= 2 - 3$$

$$= -1$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -1(x - 0)$$

$$y - 1 = -x$$

$$x + y - 1 = 0$$

$$f) A(-2, -5) \quad B(1, 4)$$

$$m = \frac{4 - (-5)}{1 - (-2)}$$

$$= \frac{9}{3}$$

$$= 3$$

$$x + 2y + 1 = 10$$

$$2y = -x - 1$$

$$y = -\frac{1}{2}x - \frac{1}{2}$$

$$\therefore m = -\frac{1}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{3}{4} + \frac{1}{2}}{1 + \frac{3}{4} \cdot \frac{1}{2}} \right|$$

$$= 7$$

$$\theta = 81^\circ 52'$$

QUESTION 2

$$a) \log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y$$

$$= 0.64 - 0.04$$

$$= 0.6$$

$$b) Q(-1, 4) \quad R(x, y) \quad P(14, -6) \quad 5:3$$

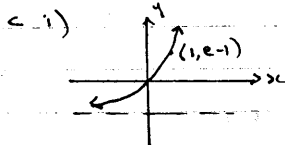
$$14 = \frac{5x - 3(-1)}{2} \quad -6 = \frac{5y - 3(4)}{2}$$

$$28 = 5x + 3 \quad -12 = 5y - 12$$

$$25 = 5x \quad 0 = 5y$$

$$x = 5 \quad y = 0$$

$$\therefore R(5, 0)$$



$$ii) V_{\text{area}} = \pi \int_a^b y^2 dx$$

$$= \pi \int_0^1 (e^x - 1)^2 dx$$

$$= \pi \int_0^1 (e^{2x} - 2e^x + 1) dx$$

$$= \pi \left[\frac{e^{2x}}{2} - 2e^x + x \right]_0^1$$

$$= \pi \left[\left(\frac{e^2}{2} - 2e + 1 \right) - \left(\frac{e^0}{2} - 2e^0 + 0 \right) \right]$$

$$= \pi \left(\frac{e^2}{2} - 2e + 2 \right) \text{ units}^3$$

$$d) i) f(x) = (x)^3 - 2(x) - 5$$

$$= -1$$

$$f(2 \cdot 2) = (2 \cdot 2)^3 - 2(2 \cdot 2) - 5$$

$$= 1 \cdot 2 + 8$$

Since one result is positive & the other negative, the root lies between 2 and 2.2.

$$ii) x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f(2 \cdot 1) = (2 \cdot 1)^3 - 2(2 \cdot 1) - 5$$

$$= 0.061$$

$$f'(x) = 3x^2 - 2$$

$$f'(2 \cdot 1) = 3(2 \cdot 1)^2 - 2$$

$$= 11.23$$

$$x_1 = 2 \cdot 1 - \frac{0.061}{11.23}$$

$$= 2.0945681$$

$$= 2.09 \text{ to 2 dec pl.}$$

QUESTION 3

$$a) \int \frac{x}{x^2+5} dx = \frac{1}{2} \int \frac{2x}{x^2+5} dx$$

$$= \frac{1}{2} \ln(x^2+5) + C$$

$$b) \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin 75 = \sin(45+30) = \sin 45 \cos 30 + \cos 45 \sin 30$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$c) \frac{4}{x+1} \cdot x(x+1)^2 \leq 3(x+1)^2$$

$$4x(x+1) \leq 3(x+1)^2$$

$$0 \leq 3(x+1)^2 - 4x(x+1)$$

$$0 \leq (x+1)(3(x+1) - 4x)$$

$$(x+1)(3x-1) \geq 0$$

$$x \leq -1 \text{ and } x \geq \frac{1}{3}$$

$$\text{but } x \neq -1$$

$$\therefore x < -1, x \geq \frac{1}{3}$$

$$d) i) f(x) = 2x^3 + 7x^2 - 7x - 12$$

$$f(-1) = 2(-1)^3 + 7(-1)^2 - 7(-1) - 12$$

$$= -2 + 7 + 7 - 12$$

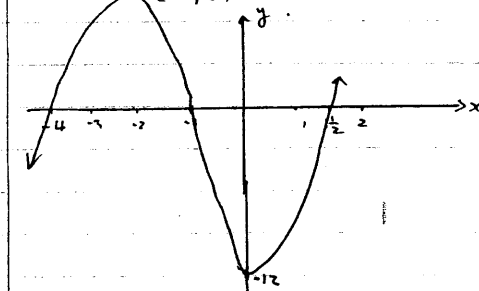
$$= 0$$

Since the remainder is 0, $(x+1)$ is a factor

$$ii) \begin{array}{r} 2x^2 + 5x - 12 \\ x+1 \overline{) 2x^3 + 7x^2 - 7x - 12} \\ \underline{2x^3 + 2x^2} \\ 5x^2 - 7x \\ \underline{5x^2 + 5x} \\ -12x - 12 \\ \underline{-12x - 12} \\ 0 \end{array}$$

$$\therefore f(x) = (x+1)(2x^2 + 5x - 12)$$

$$= (x+1)(2x-3)(x+4) \quad x = -1, \frac{3}{2}, -4$$



QUESTION 4

$$\alpha + \beta + \gamma = \frac{-b}{a} = \frac{6}{3} = 2$$

$$\alpha + \beta + \gamma = \frac{a}{a} = \frac{1}{1} = 1$$

$$\alpha + \beta = -\frac{a}{a} = -\frac{2}{3}$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{1}{-\frac{2}{3}}$$

$$= -\frac{1}{2}$$

$$i) y = \frac{x^2}{4}$$

$$y' = \frac{2x}{4} = \frac{x}{2}$$

$$\text{at } x=2t, y' = \frac{2t}{2} = t$$

$$y - y_1 = m(x - x_1)$$

$$y - t^2 = t(x - 2t)$$

$$y - t^2 = tx - 2t^2$$

$$0 = tx - y - t^2$$

$$ii) \text{ If } y=0, tx - 0 - t^2 = 0$$

$$x = \frac{t^2}{t} = t$$

$$\therefore T(t, 0)$$

$$\text{midpoint} = \left(\frac{\frac{t^2}{2} + t, \frac{t^2 + 0}{2}} \right) = \left(\frac{3t}{2}, \frac{t^2}{2} \right)$$

$$ii) x = \frac{3t}{2}, y = \frac{t^2}{2}$$

$$t = \frac{2x}{3}$$

$$y = \frac{4x^2}{9} \times \frac{1}{2}$$

$$18y = 4x^2$$

$$9y = 2x^2$$

$$c) i) \angle ADC = 90^\circ \text{ angle in a semi-circle}$$

$$ii) \angle DCA = 180 - 90 - 52 \text{ (angle sum of } \Delta)$$

$$= 38^\circ$$

$$\angle DBA = 38^\circ \text{ (angles in the same segment are equal)}$$

QUESTION 5

$$a) 3\% \text{ pa} = \frac{1}{4}\% \text{ pm} = 0.0025$$

$$i) A = P(1+r)^n$$

$$= 150(1.0025)^{72}$$

$$= \$179.54$$

$$ii) A_1 = 150(1.0025)^{72}$$

$$A_2 = 150(1.0025)^{71}$$

$$\vdots$$

$$A_{72} = 150(1.0025)^1$$

$$\text{Total } A = 150(1.0025 + 1.0025^2 + \dots + 1.0025^{72})$$

$$= 150 \left(\frac{1.0025(1.0025^{72} - 1)}{1.0025 - 1} \right)$$

$$= \$11846.45$$

$$b) \text{ If } n=1, \text{ LHS} = \frac{1}{2 \times 3} = \frac{1}{6}$$

$$\text{RHS} = \frac{1}{2(3)} = \frac{1}{6} = \text{LHS}$$

$$\therefore \text{ true for } n=1$$

$$\text{Assume true for } n=k$$

$$\text{i.e. } \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(k+1)(k+2)} = \frac{k}{2(k+2)}$$

$$\text{To prove true for } n=k+1$$

$$\text{i.e. } \frac{1}{2 \times 3} + \dots + \frac{1}{(k+1)(k+2)} + \frac{1}{(k+2)(k+3)} = \frac{k+1}{2(k+3)}$$

$$\text{LHS} = \frac{k}{2(k+2)} + \frac{1}{(k+2)(k+3)}$$

$$= \frac{k(k+3) + 1(2)}{2(k+2)(k+3)}$$

$$= \frac{k^2 + 3k + 2}{2(k+2)(k+3)}$$

$$= \frac{(k+2)(k+1)}{2(k+2)(k+3)}$$

$$= \frac{k+1}{2(k+3)}$$

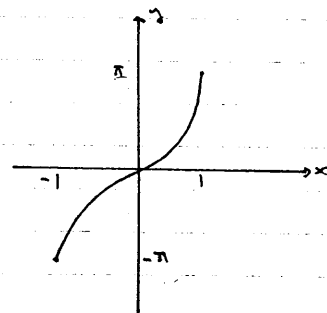
$$\therefore \text{ true for } n=k+1$$

$$\text{Since statement is true for } n=1$$

$$\text{it is true for } n=1+1=2, \text{ and } n=2+1=3$$

$$\text{etc.}$$

$$c) i)$$



$$ii) f\left(\frac{1}{\sqrt{2}}\right) = 2 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$= 2 \cdot \frac{\pi}{4}$$

$$= \frac{\pi}{2}$$

$$iii) f(x) = 2 \sin^{-1} x$$

$$f'(x) = 2 \cdot \frac{1}{\sqrt{1-x^2}}$$

$$f'\left(\frac{1}{\sqrt{2}}\right) = \frac{2}{\sqrt{1-\frac{1}{2}}}$$

$$= \frac{2}{\frac{1}{\sqrt{2}}}$$

$$= 2 \div \frac{1}{\sqrt{2}}$$

$$= 2\sqrt{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{\pi}{2} = 2\sqrt{2}\left(x - \frac{1}{\sqrt{2}}\right)$$

$$y - \frac{\pi}{2} = 2\sqrt{2}x - 2$$

$$y = 2\sqrt{2}x - 2 + \frac{\pi}{2}$$

QUESTION 6

$$a) \frac{d}{dx} \cos^{-1}(2x^3) = \frac{-1}{\sqrt{1-(2x^3)^2}} \cdot 6x^2$$

$$= \frac{-6x^2}{\sqrt{1-4x^6}}$$

$$b) \cos \theta (\tan \theta - \tan \frac{1}{2} \theta)$$

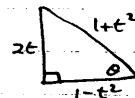
$$= \frac{1-t^2}{1+t^2} \left(\frac{2t}{1-t^2} - t \right)$$

$$= \frac{1-t^2}{1+t^2} \left(\frac{2t - t + t^3}{1-t^2} \right)$$

$$= \frac{t + t^3}{1+t^2}$$

$$= \frac{t(1+t^2)}{1+t^2}$$

$$= t = \tan \frac{1}{2} \theta$$



$$c) \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \cos^2 x \, dx =$$

$$\frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} (\cos 2x + 1) \, dx$$

$$= \frac{1}{2} \left[\frac{1}{2} \sin 2x + x \right]_{\frac{\pi}{2}}^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left[\left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin \pi + \frac{\pi}{4} \right) \right]$$

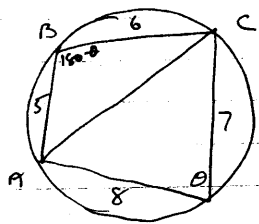
$$= \frac{1}{2} \left[\left(0 + \frac{\pi}{2} \right) - \left(\frac{1}{2} + \frac{\pi}{4} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$= \frac{\pi}{8} - \frac{1}{4}$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 2\cos^2 x - 1 \\ \cos 2x + 1 &= 2\cos^2 x \end{aligned}$$

1) $\angle ABC = 180^\circ - \theta$ opposite $\angle$ s of cyclic quadrilateral are supplementary.



$$\cos \theta = \frac{8^2 + 7^2 - AC^2}{2 \cdot 8 \cdot 7}$$

$$= \frac{113 - AC^2}{112}$$

$$AC^2 = 5^2 + 6^2 - 2 \cdot 5 \cdot 6 \cdot \cos(180^\circ - \theta)$$

$$= 61 + 60 \cos \theta$$

$$\therefore \cos \theta = \frac{113 - (61 + 60 \cos \theta)}{112}$$

$$112 \cos \theta = 113 - 61 - 60 \cos \theta$$

$$172 \cos \theta = 52$$

$$\cos \theta = \frac{52}{172}$$

$$= \frac{13}{43}$$

QUESTION 7

i) $u = e^x$

$$\frac{du}{dx} = e^x$$

$$dx = \frac{du}{e^x}$$

$$\int \frac{e^x dx}{1+e^{2x}} = \int \frac{\frac{1}{u}}{1+u^2} \cdot \frac{du}{u}$$

$$= \int \frac{du}{1+u^2}$$

$$= \tan^{-1} u + C$$

$$= \tan^{-1} e^x + C$$

b) i) $R = \sqrt{a^2 + b^2}$ $a = \sqrt{3}$, $b = 1$

$$R = \sqrt{(\sqrt{3})^2 + 1^2}$$

$$= 2$$

$$\tan \alpha = \frac{b}{a} \quad 0 < \alpha < \frac{\pi}{2}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = 30^\circ = \frac{\pi}{6}$$

$$R \sin(\theta - \alpha) = 2 \sin(\theta - \frac{\pi}{6})$$

ii) $2 \sin(\theta - \frac{\pi}{6}) = 1$

$$\sin(\theta - \frac{\pi}{6}) = \frac{1}{2}$$

$$\theta - \frac{\pi}{6} = \frac{\pi}{6}, \quad \theta - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\theta = \frac{\pi}{3}$$

$$\theta = \pi$$

c) i) $f(x) = \sqrt{x+6}$

$$y = \sqrt{x+6}$$

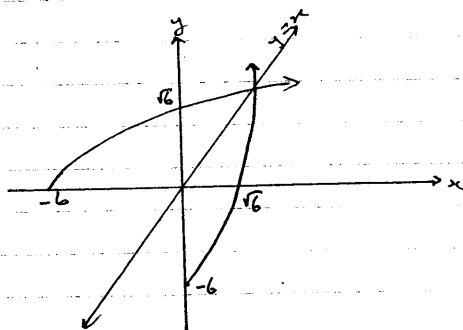
$$x = y^2 + 6$$

$$x^2 = y^2 + 6$$

$$y = x^2 - 6$$

$$f^{-1}(x) = x^2 - 6$$

ii)



iii) $D: x \geq 0$

iv)

$y = x^2 - 6$ intersects with $y = x$.

$$x = x^2 - 6$$

$$x^2 - x - 6 = 0$$

$y = \sqrt{x+6}$ intersects with $y = x$

$$x = \sqrt{x+6}$$

$$x^2 = x+6$$

$$x^2 - x - 6 = 0$$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$x = 3, -2$$

but $x \neq -2$

$$\therefore x = 3$$

Point of intersection is (3, 3)